HMO Market Data Analytics Proof of Concept

India VIX Historical Market Data | Power BI Analytics Project

Project Report

1. Project Objective

The objective of this Proof of Concept is to demonstrate an end-to-end Data Analytics workflow using real historical Indian financial-market data. The project uses daily India VIX data and covers data understanding, cleaning and preparation, analytical calculations, DAX measures, interactive Power BI dashboards, and data-driven insights.

2. Data Source & Understanding

The working dataset used for the project was obtained from Kaggle, as specified during the project setup. It contains historical India VIX observations. The dashboard covers 1,627 records from 02 January 2020 to 24 August 2026. The primary market fields are Date, Close, High, Low, Open and Volume.

For source validation, the National Stock Exchange of India (NSE) provides an official Historical Data – India VIX page. NSE is the authoritative exchange source for India VIX historical reporting. The exact Kaggle dataset URL should be copied from the original Kaggle download page used for the submitted CSV; it is intentionally not fabricated here.

Official NSE reference: https://www.nseindia.com/reports-indices-historical-vix

Kaggle acquisition source: User-provided Kaggle India VIX historical dataset.

3. Dataset Scope

Item	Value
Rows	1,627 daily observations
Date range	02-Jan-2020 to 24-Aug-2026
Core fields	Date, Close, High, Low, Open, Volume
Analysis focus	India VIX level, daily movement, intraday range, yearly/monthly patterns

4. Data Dictionary

Field	Type	Definition	Project Use
Date	Date	Trading date of the India VIX observation.	Time-series analysis, slicers, drill-through
Open	Numeric	Opening India VIX value for the trading session.	OHLC analysis
High	Numeric	Highest India VIX value recorded during the session.	Peak and intraday analysis

Low	Numeric	Lowest India VIX value recorded during the session.	Intraday analysis
Close	Numeric	Closing India VIX value for the trading session.	Primary VIX trend and KPI analysis
Volume	Numeric	Volume field supplied in the source dataset.	Retained for completeness; not central to this POC
Previous Close	Numeric	Prior trading-session closing VIX derived from Close.	Daily movement calculation
Daily Change	Numeric	Close minus Previous Close.	Absolute daily movement
Daily Return %	Numeric (%)	Percentage change from Previous Close to Close.	Relative daily movement
High-Low Range	Numeric	High minus Low.	Intraday volatility proxy

5. Data Cleaning & Preparation

- Loaded the historical India VIX dataset into Power BI Power Query.
- Inspected column names, data types, validity indicators, missing values and error counts.
- Converted Date to a proper date type and numeric market fields to numeric data types.
- Checked for duplicate observations using the trading date as the key business field.
- Handled the first-row calculation limitation for Previous Close / Daily Return by allowing the first observation to remain null where no prior trading session exists.
- Created Previous Close from the prior trading-session Close.
- Created Daily Change = Close - Previous Close.
- Created Daily Return % = (Close - Previous Close) / Previous Close × 100.
- Created High-Low Range = High - Low.
- Created a DateTable containing Date, Year, Month Number, Month Name, Year Month and related sorting fields.
- Created a relationship between DateTable and the India VIX fact table to support time filtering and DAX calculations.

Note: The first record has no previous trading-day observation, so a null/blank Daily Return for that row is expected and is not treated as a data-quality error.

6. Analysis Performed

- Current, average, maximum and minimum India VIX KPIs.
- Historical closing VIX trend from 2020 through August 2026.
- Year-wise average VIX and peak VIX comparison.
- Daily VIX percentage-return and absolute-change analysis.
- Intraday High-Low range analysis.
- Positive versus negative VIX movement days by year.
- Volatility spike-day count using an analytical threshold of |Daily Return %| ≥ 10%.
- Year-month heatmap of average India VIX.
- Calendar-month average VIX comparison.

- Drill-through analysis for individual trading sessions.

7. Power BI Dashboard Structure

Page	Purpose
01_Executive Overview	Executive KPIs, closing trend, year comparison, market highlights and date/year filters.
02_Volatility Deep Dive	Daily returns, intraday range, movement-day comparison, volatility spike KPI and largest positive movements table.
03_Time Analysis	Yearly trend, monthly trend, calendar-month comparison and Year × Month heatmap.
04_Daily Details	Drill-through page for individual trading-session details.

8. Important DAX Measures

Measure	Purpose
Current VIX	Latest available closing VIX based on the maximum selected date.
Average VIX	Average closing India VIX over the selected filter context.
Maximum VIX	Highest closing India VIX in the selected context.
Minimum VIX	Lowest closing India VIX in the selected context.
Trading Days	Distinct count of trading dates.
Average Daily Change	Average absolute day-to-day VIX movement.
Average Daily Return	Average daily percentage movement.
Maximum Daily Increase	Largest positive Daily Return %.
Maximum Daily Decrease	Most negative Daily Return %.
Average Intraday Range	Average High-Low range.
Volatility Spike Days	Count of days with absolute Daily Return % ≥ 10%.
Positive Movement Days	Count of days where Daily Change > 0.
Negative Movement Days	Count of days where Daily Change < 0.
Highest VIX Date	Date corresponding to the highest closing VIX.
Lowest VIX Date	Date corresponding to the lowest closing VIX.
Largest Daily Increase Date	Date corresponding to the largest positive Daily Return %.

9. Key Findings & Insights

- The latest available observation in the dashboard is 24-Aug-2026, with a closing India VIX of 11.52.
- The average closing India VIX across the analysed period is 17.27. The latest value of 11.52 is approximately 33.3% below this historical average, indicating a comparatively calmer volatility level at the end of the selected period.
- The highest closing VIX in the working dataset is 83.61, recorded on 24-Mar-2020. This is the dominant peak visible in the historical trend and is concentrated in the 2020 period.
- The lowest closing VIX in the working dataset is 9.15, recorded on 26-Dec-2025.
- 2020 has the highest annual average VIX at approximately 27, while 2023 has the lowest annual average at approximately 12 among the years displayed.

- The largest positive one-day VIX movement in the dashboard is 65.63% on 07-Apr-2025. The largest negative daily movement is approximately -29.38%.
- The dashboard identifies 92 volatility-spike days using the project-defined threshold of absolute Daily Return % ≥ 10%. This threshold is an analytical rule used for this POC, not an official India VIX classification.
- The calendar-month comparison shows the highest average VIX in April (approximately 23.7) followed by March (approximately 22.1) and June (approximately 21.0) in the displayed analysis. The lowest monthly average shown is around September (approximately 14.8).
- The Year × Month heatmap shows that elevated volatility is concentrated in particular periods rather than being uniformly high across all years and months.

10. Challenges Faced

- The first trading observation cannot have a Previous Close, so the first derived return is naturally blank.
- Date handling required a dedicated DateTable to support reliable yearly/monthly analysis and slicers.
- Power BI aggregation choices had to be controlled so that VIX values were averaged or maximized appropriately rather than unintentionally summed.
- Dashboard layout required balancing analytical depth with readability across multiple pages.
- The project uses a Kaggle acquisition source, while NSE is the authoritative market-data reference; the exact Kaggle dataset URL should be preserved with the final source files.

11. Final Conclusion

This Proof of Concept demonstrates a complete Power BI data-analytics workflow on real Indian financial-market data. The analysis transforms raw India VIX observations into a structured analytical model, derives daily movement and intraday-range metrics, and presents the results through interactive dashboards, slicers and drill-through functionality. The results show clear variation in volatility across years and months, with a pronounced peak in 2020 and materially lower volatility in several later periods. The solution provides a reusable foundation for future market-risk or financial-data analytics work.